EXPERIMENT TO UNDERSTAND EDA QUANTITATIVE AND QUALITATIVE ANALYSIS

PROGRAM:

```
import warnings
warnings.filterwarnings("ignore")

import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

%matplotlib inline

tips = sns.load_dataset('tips')

tips.head()

sns.displot(tips.total_bill, kde=True)
plt.show()

sns.displot(tips.total_bill, kde=False)
plt.show()

sns.jointplot(x=tips.tip, y=tips.total_bill)
plt.show()
```

```
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
plt.show()

sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")
plt.show()

sns.pairplot(tips)
plt.show()

tips.time.value_counts()

sns.pairplot(tips, hue='time')
plt.show()

sns.pairplot(tips, hue='day')
plt.show()

sns.heatmap(tips.corr(numeric_only=True), annot=True)
plt.show()

sns.boxplot(tips.total_bill)
plt.show()

sns.boxplot(tips.tip)
plt.show()
```

```
sns.countplot(tips.day)
plt.show()

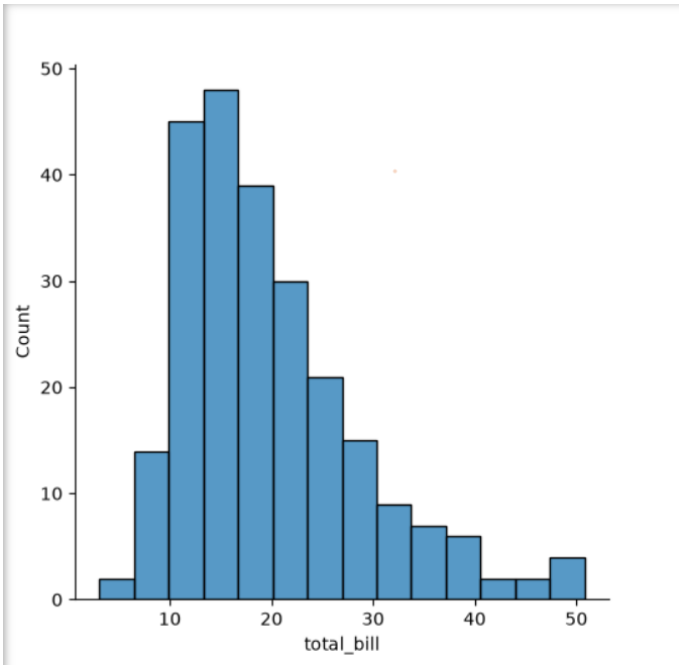
sns.countplot(tips.sex)
plt.show()

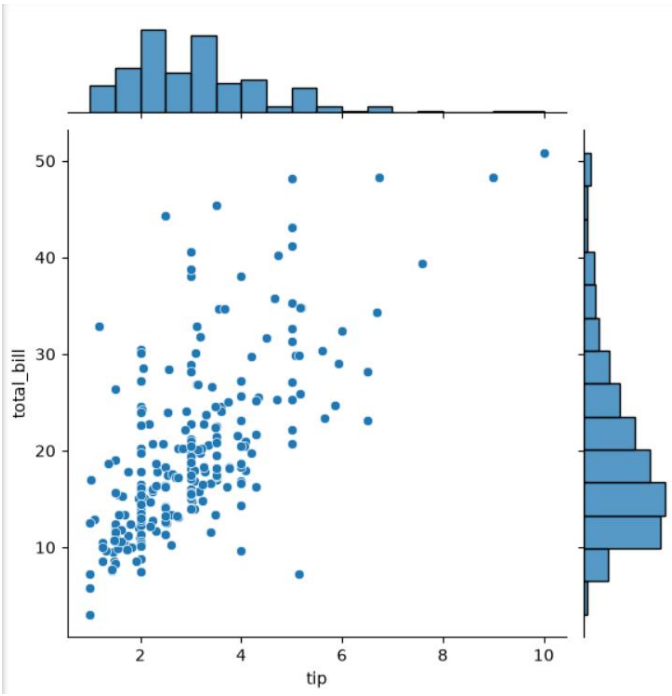
tips.sex.value_counts().plot(kind='pie')
plt.show()

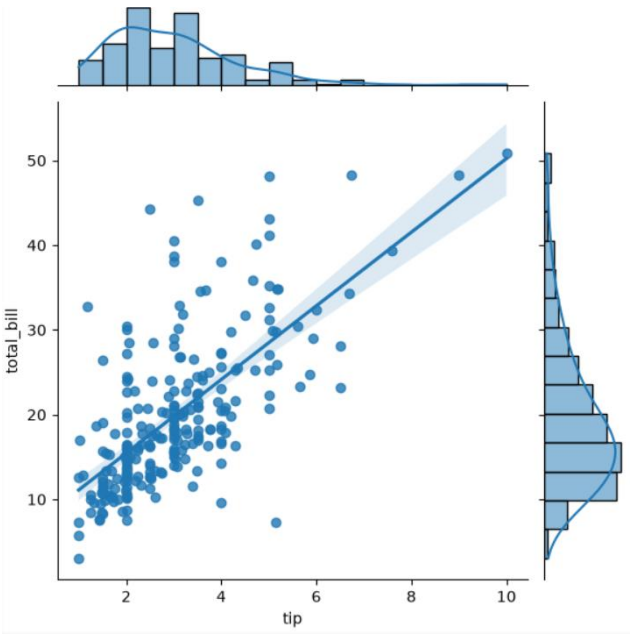
tips.sex.value_counts().plot(kind='bar')
plt.show()

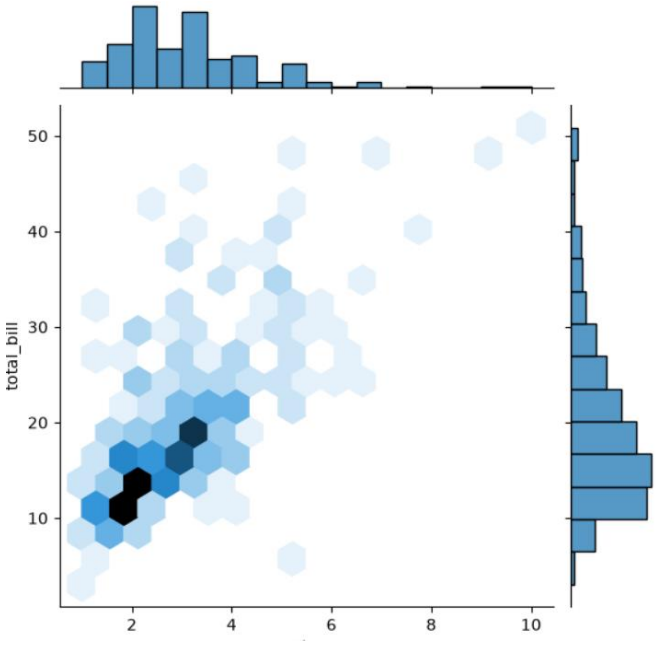
sns.countplot(tips[tips.time == 'Dinner']['day'])
plt.show()
```

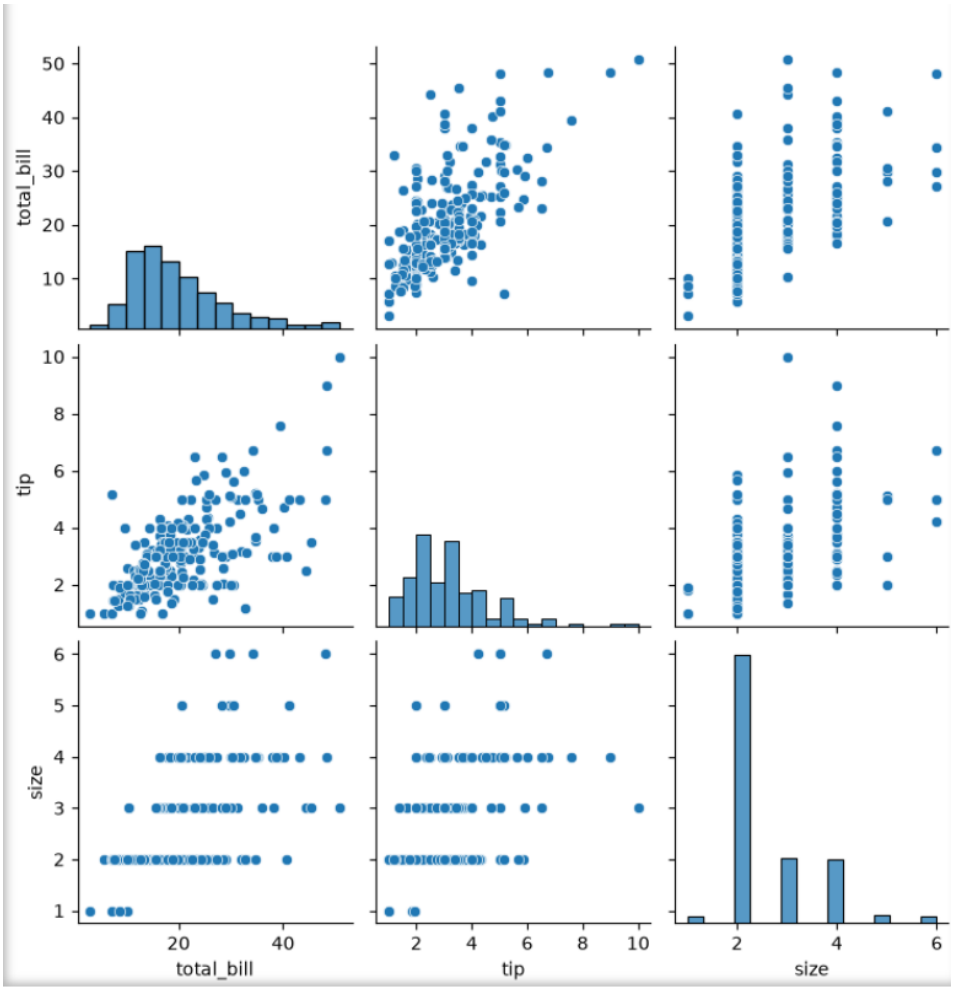
Output:

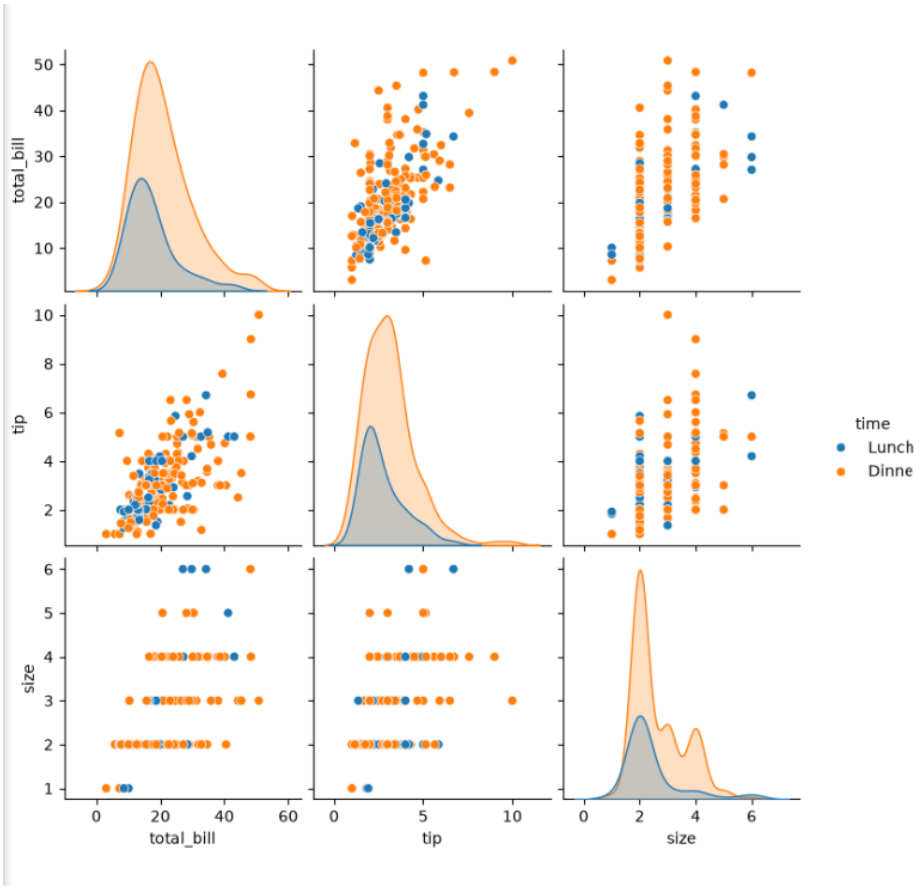


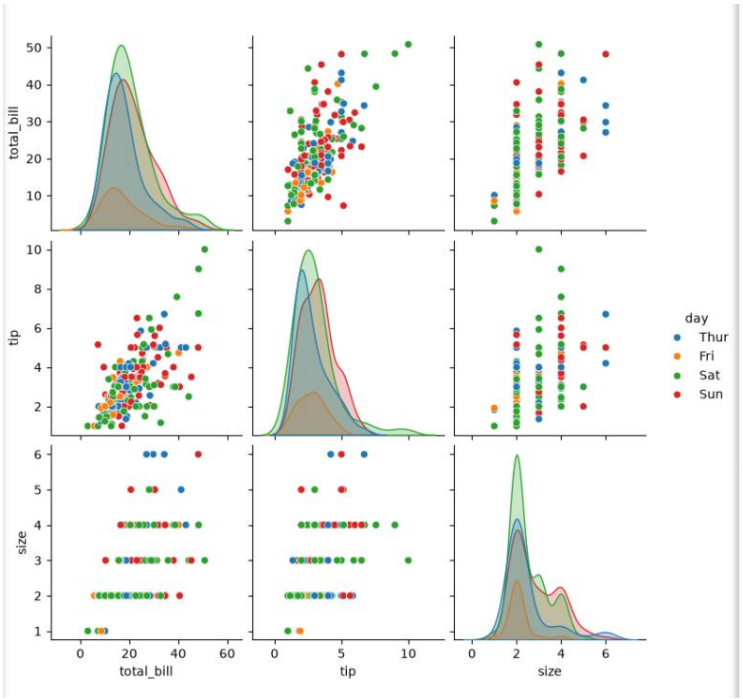

total_bill
tip
50
40
30
20
10
2
4
6
8
10

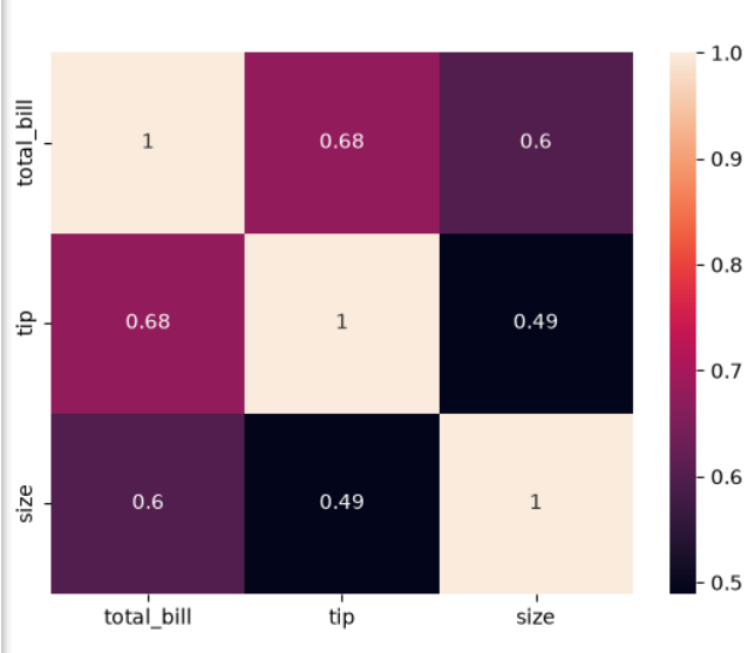

total_bill
tip
50
40
30
20
10
2
4
6
8
10

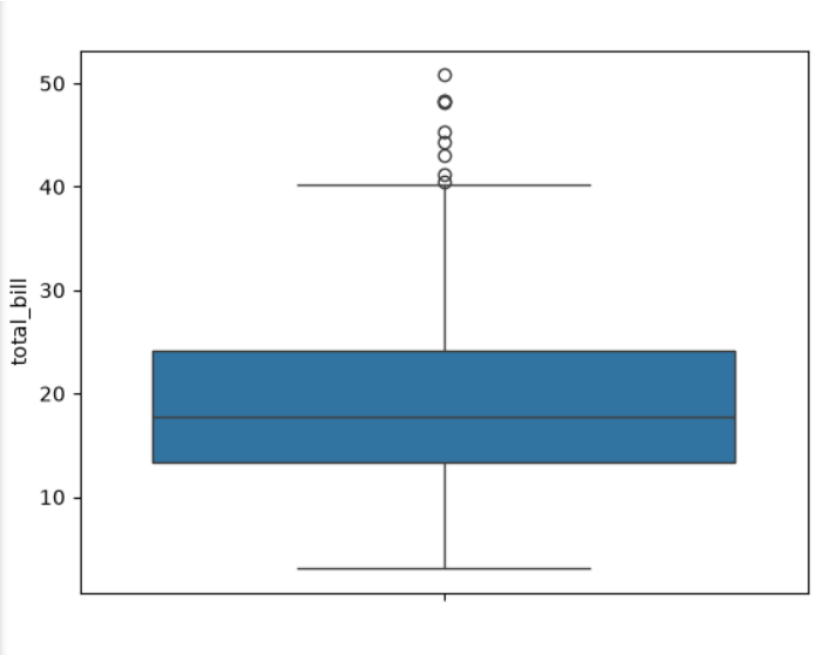

total_bill
50
40
30
20
10
2
4
6
8
10

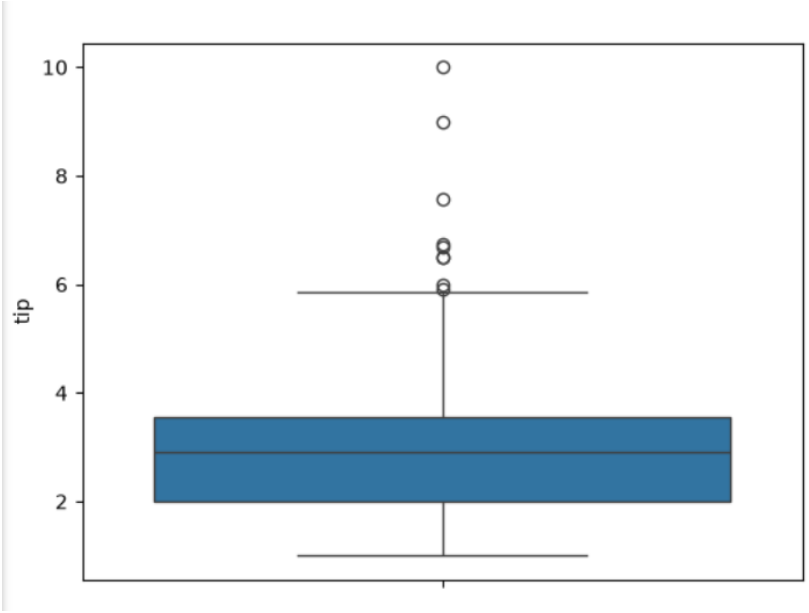

total_bill
tip
size
50
40
30
20
10
10
8
6
4
2
6
5
4
3
2
1
20
40
2
4
6
8
10
2
4
6

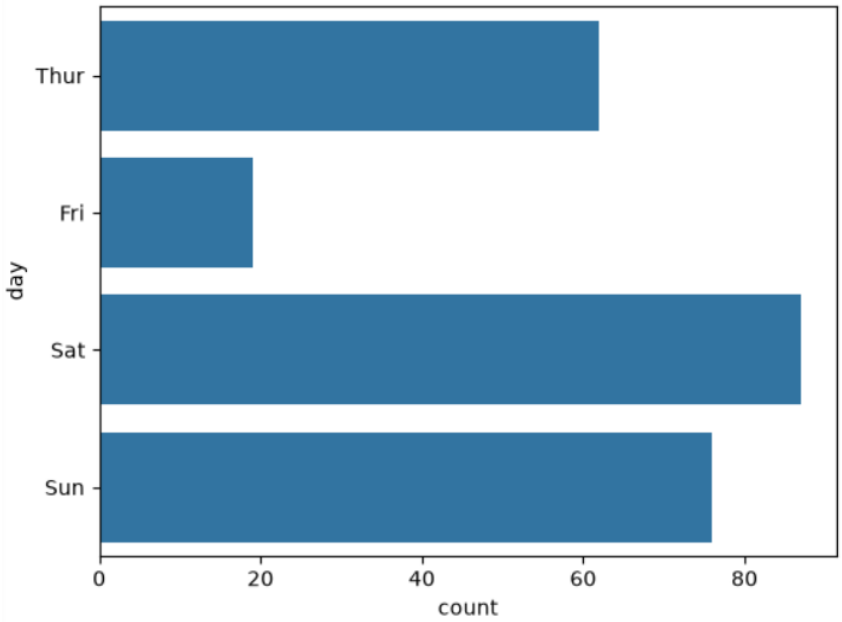

total_bill
tip
size
total_bill
tip
size
time
Lunch
Dinne

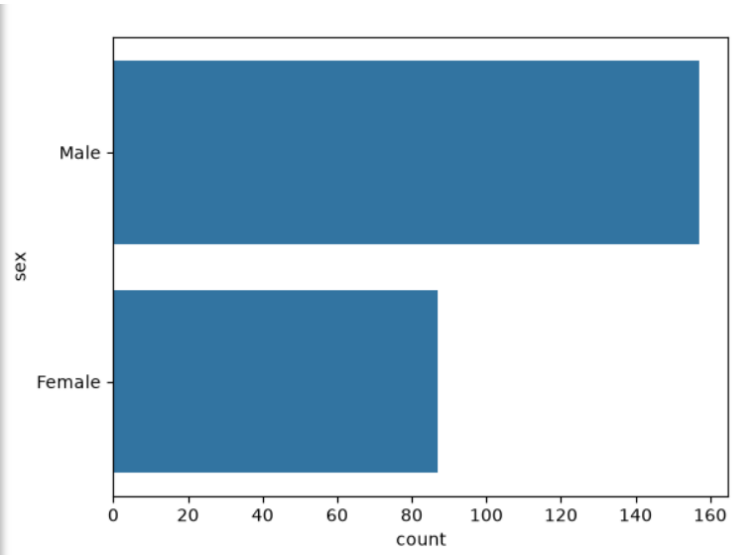

total_bill
tip
size
total_bill
tip
size
day
Thur
Fri
Sat
Sun

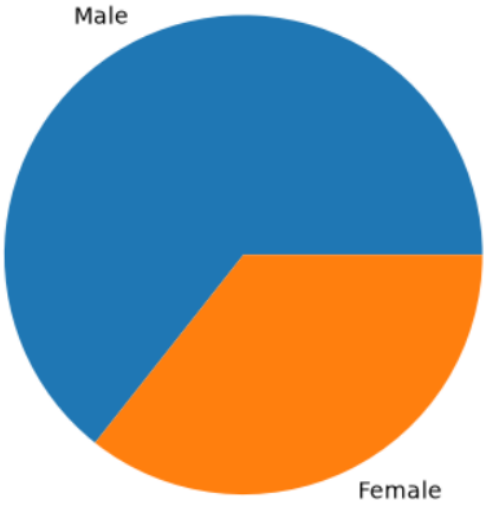

total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
total_bill
tip
size
1.0
0.9
0.8
0.7
0.6
0.5

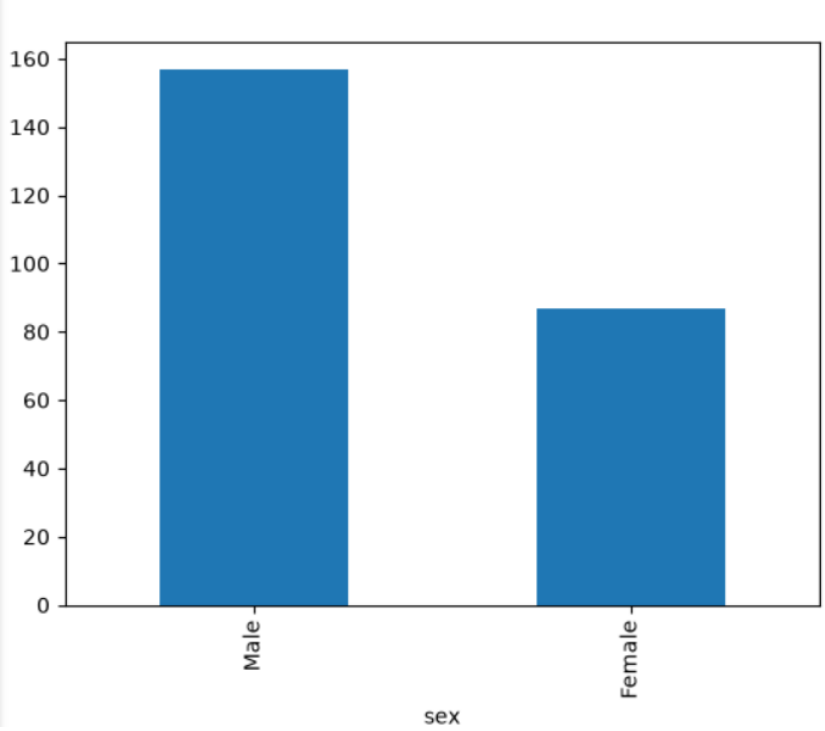

total_bill
50
40
30
20
10

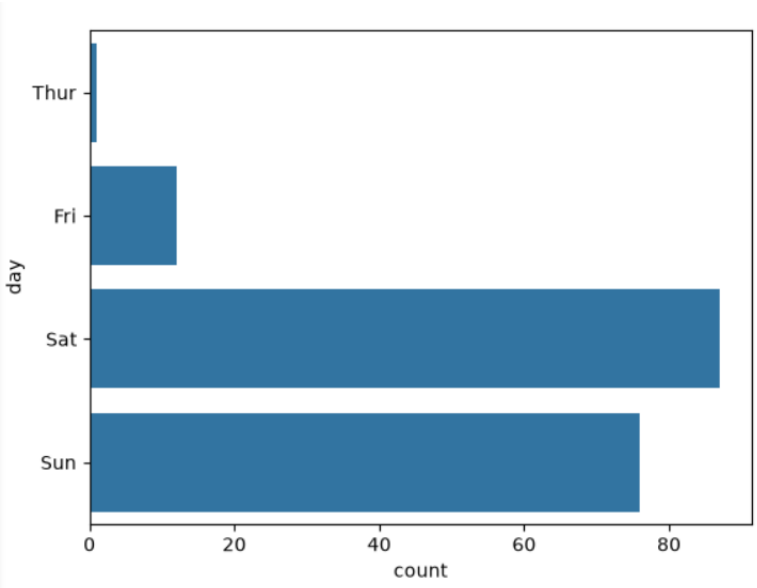

10
8
6
4
2
tip


Thur
Fri
Sat
Sun
day
0
20
40
60
80
count


Male
Female
sex
0
20
40
60
80
100
120
140
160
count


Male
Female


160
140
120
100
80
60
40
20
0
Male
Female
sex


Thur
Fri
Sat
Sun
day
0
20
40
60
80
count